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ELECTRONIC DEVICE FOR DETERMINING IMAGE VIEW AREA AND OPERATION METHOD THEREFOR

Abstract

An electronic device is provided. The electronic device includes a camera module, a display, memory, comprising one or more storage media, storing instructions, and at least one processor operatively connected with the camera module, the display, and the memory, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to identify a maximum view area that is provided by the camera module in response to identifying that an auto-framing function is activated while a camera application is being executed, acquire a first user input for designating a movable view area range within the maximum view area, acquire a second user input for designating an object area to be tracked within the view area range, determine a first view area including the object area on the basis of the first user input and the second user input, and display an image corresponding to the determined first view area on the display.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365(c), of an International application No. PCT/KR2023/017777, filed on Nov. 7, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0161299, filed on Nov. 28, 2022, in the Korean Intellectual Property Office, and of a Korean application number 10-2023-0003536, filed on Jan. 10, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to an electronic device for determining a view area of an image and an operation method therefor.

2. Description of Related Art

[0003] As carrying portable electronic devices has been generalized in our daily lives, the use of camera functions of portable electronic devices has increased significantly, and photographing with a portable electronic device is recognized as an indispensable function to the extent that the camera functions become a criterion for selecting a portable electronic device. Recently, as the camera functions implemented in portable electronic devices have been enhanced, auto-framing for automatically optimizing a photographing frame according to a subject and adjusting a composition has rapidly become a standard function of portable electronic devices. For example, when the auto-framing function is activated, an electronic device may detect subjects to photograph, and may provide an image by adjusting a photographing angle and/or a zoom according to the number, motions, or positions of detected subjects.

[0004] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0005] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device for determining image view area and operation method therefor.

[0006] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0007] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a camera module, a display, memory, comprising one or more storage media, storing instructions, and at least one processor operatively connected with the camera

module, the display, and the memory operatively connected with the at least one processor, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to identify a maximum view area that is provided by the camera module in response to identifying that an auto-framing function is activated while a camera application is being executed, acquire a first user input for designating a view area range that is movable within the maximum view area, acquire a second user input for designating an object area to track within the view area range, determine a first view area including the object area based on the first user input and the second user input, and display an image corresponding to the determined first view area on the display.

[0008] In accordance with an aspect of the disclosure, an operation method of an electronic device is provided. The method includes identifying a maximum view area that is provided by a camera module of the electronic device in response to identifying that an auto-framing function is activated while a camera application is being executed, acquiring a first user input for designating a view area range that is movable within the maximum view area, acquiring a second user input for designating an object area to track within the view area range, determining a first view area comprising the object area based on the first user input and the second user input, and displaying an image corresponding to the determined first view area.

[0009] In accordance with another aspect of the disclosure, one or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by one or more processors of an electronic device individually or collectively, cause the electronic device to perform operations are provided. The operations include identifying a maximum view area that is provided by a camera module of the electronic device in response to identifying that an auto-framing function is activated while a camera application is being executed, acquiring a first user input for designating a view area range that is movable within the maximum view area, acquiring a second user input for designating an object area to track within the view area range, determining a first view area comprising the object area based on the first user input and the second user input, and displaying an image corresponding to the determined first view area.

[0010] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0012] FIG. 1 is a block diagram of an electronic device in a network environment according to an embodiment of the disclosure;

[0013] FIG. 2 is a block diagram illustrating a camera module according to an embodiment of the disclosure;

[0014] FIG. 3 is a block diagram of an electronic device according to an embodiment of the disclosure;

[0015] FIGS. 4A, 4B, 4C, and 4D are views illustrating a method of determining a view area and displaying when an auto-framing function is activated according to various embodiments of the disclosure;

[0016] FIG. 5 is a flowchart illustrating an operation method of an electronic device according to an embodiment of the disclosure;

[0017] FIG. 6 is a flowchart to explain detailed operations of determining a view area range by an electronic device according to an embodiment of the disclosure;

[0018] FIG. 7 is a flowchart to explain detailed operations of determining an object area by an electronic device according to an embodiment of the disclosure;

[0019] FIG. 8 is a view to explain a method for providing an image corresponding to a view area determined based on an object area according to an embodiment of the disclosure;

[0020] FIG. 9 is a view to explain a method for providing an image corresponding to a view area determined based on an object area according to an embodiment of the disclosure;

[0021] FIG. 10 is a view to explain a method for providing an image corresponding to a view area determined based on an object area according to an embodiment of the disclosure; and

[0022] FIG. 11 is a view to explain a method for providing an image corresponding to a view area determined based on a plurality of object areas according to an embodiment of the disclosure.

[0023] Throughout the drawings, like reference numerals will be understood to refer to like parts, components, and structures.

DETAILED DESCRIPTION

[0024] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0025] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0026] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0027] The auto-framing function of a camera may refer to a function of tracking a main subject positioned within a maximum view area that may be obtained by the camera, focusing on the main subject, and adjusting a photographing composition to continuously center the main subject in an image frame. When the auto-framing function is activated, the electronic device may automatically determine the photographing composition with reference to a subject recognized within the maximum view area of the camera. Accordingly, a moving range of a view area that is seen through the camera when the auto-framing function is activated may be fixed within the maximum view area of the camera regardless of a size or composition of the subject desired by the user. In addition, when a setting for tracking the subject during photographing is disabled, the electronic device may re-detect the subject based on the maximum view area of the camera and may re-configure the photographing composition, but in this process, the view area that is seen through the camera may be greatly changed, causing the user inconvenience.

[0028] In various embodiments of the disclosure, when the auto-framing function of a camera is used, a user may be allowed to designate a movable view area range within a maximum view area of the camera, and an area of a subject to track within the view area range, so that an image of a composition desired by the user may be provided. In various embodiments of the disclosure, an area range for detecting a subject within a maximum view area of the camera and an area to be designated as a subject may be recommended based on image analysis, and setting by the user may be induced, so that it is possible to take images that meet user's needs. In various embodiments of

the disclosure, by tracking a subject and changing a view area based on a designated view area range within a maximum view area of the camera, unnecessary shift of a view area may be prevented, and calculation related to the auto-framing function may be rapidly and stably processed.

[0029] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0030] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless fidelity (Wi-Fi) chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a fingerprint sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0031] FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to an embodiment of the disclosure.

[0032] Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0033] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main

processor **121**.

[0034] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0035] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0036] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0037] The input module **150** may receive a command or data to be used by other component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0038] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0039] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0040] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0041] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state.

According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0042] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0043] The connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0044] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0045] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0046] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0047] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0048] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a fifth-generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0049] The wireless communication module **192** may support a 5G network, after a fourth-

generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the millimeter wave (mmWave) band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0050] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0051] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0052] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0053] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service

related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0054] FIG. **2** is a block diagram **200** illustrating the camera module **180** according to an embodiment of the disclosure.

[0055] Referring to FIG. **2**, the camera module **180** may include a lens assembly **210**, a flash **220**, an image sensor **230**, an image stabilizer **240**, memory **250** (e.g., buffer memory), or an image signal processor **260**. The lens assembly **210** may collect light emitted or reflected from an object whose image is to be taken. The lens assembly **210** may include one or more lenses. According to an embodiment, the camera module **180** may include a plurality of lens assemblies **210**. In such a case, the camera module **180** may form, for example, a dual camera, a 360-degree camera, or a spherical camera. Some of the plurality of lens assemblies **210** may have the same lens attribute (e.g., view angle, focal length, auto-focusing, f number, or optical zoom), or at least one lens assembly may have one or more lens attributes different from those of another lens assembly. The lens assembly **210** may include, for example, a wide-angle lens or a telephoto lens.

[0056] The flash **220** may emit light that is used to reinforce light reflected from an object. According to an embodiment, the flash **220** may include one or more light emitting diodes (LEDs) (e.g., a red-green-blue (RGB) LED, a white LED, an infrared (IR) LED, or an ultraviolet (UV) LED) or a xenon lamp. The image sensor **230** may obtain an image corresponding to an object by converting light emitted or reflected from the object and transmitted via the lens assembly **210** into an electrical signal. According to an embodiment, the image sensor **230** may include one selected from image sensors having different attributes, such as a RGB sensor, a black-and-white (BW) sensor, an IR sensor, or a UV sensor, a plurality of image sensors having the same attribute, or a plurality of image sensors having different attributes. Each image sensor included in the image sensor **230** may be implemented using, for example, a charged coupled device (CCD) sensor or a complementary metal oxide semiconductor (CMOS) sensor.

[0057] The image stabilizer **240** may move the image sensor **230** or at least one lens included in the lens assembly **210** in a particular direction, or control an operational attribute (e.g., adjust the read-out timing) of the image sensor **230** in response to the movement of the camera module **180** or the electronic device **101** including the camera module **180**. This allows compensating for at least part of a negative effect (e.g., image blurring) by the movement on an image being captured. According to an embodiment, the image stabilizer **240** may sense such a movement by the camera module **180** or the electronic device **101** using a gyro sensor (not shown) or an acceleration sensor (not shown) disposed inside or outside the camera module **180**. According to an embodiment, the image stabilizer **240** may be implemented, for example, as an optical image stabilizer. The memory **250** may store, at least temporarily, at least part of an image obtained via the image sensor **230** for a subsequent image processing task. For example, if image capturing is delayed due to shutter lag or multiple images are quickly captured, a raw image obtained (e.g., a Bayer-patterned image, a high-resolution image) may be stored in the memory **250**, and its corresponding copy image (e.g., a low-resolution image) may be previewed via the display module **160**. Thereafter, if a specified condition is met (e.g., by a user's input or system command), at least part of the raw image stored in the memory **250** may be obtained and processed, for example, by the image signal processor **260**.

According to an embodiment, the memory **250** may be configured as at least part of the memory **130** or as a separate memory that is operated independently from the memory **130**.

[0058] The image signal processor **260** may perform one or more image processing with respect to an image obtained via the image sensor **230** or an image stored in the memory **250**. The one or more image processing may include, for example, depth map generation, three-dimensional (3D) modeling, panorama generation, feature point extraction, image synthesizing, or image compensation (e.g., noise reduction, resolution adjustment, brightness adjustment, blurring, sharpening, or softening). Additionally or alternatively, the image signal processor **260** may perform control (e.g., exposure time control or read-out timing control) with respect to at least one (e.g., the image sensor **230**) of the components included in the camera module **180**. An image processed by the image signal processor **260** may be stored back in the memory **250** for further processing, or may be provided to an external component (e.g., the memory **130**, the display module **160**, the electronic device **102**, the electronic device **104**, or the server **108**) outside the camera module **180**. According to an embodiment, the image signal processor **260** may be configured as at least part of the processor **120**, or as a separate processor that is operated independently from the processor **120**. If the image signal processor **260** is configured as a separate processor from the processor **120**, at least one image processed by the image signal processor **260** may be displayed, by the processor **120**, via the display module **160** as it is or after being further processed.

[0059] According to an embodiment, the electronic device **101** may include a plurality of camera modules **180** having different attributes or functions. In such a case, at least one of the plurality of camera modules **180** may form, for example, a wide-angle camera and at least another of the plurality of camera modules **180** may form a telephoto camera. Similarly, at least one of the plurality of camera modules **180** may form, for example, a front camera and at least another of the plurality of camera modules **180** may form a rear camera.

[0060] FIG. **3** is a block diagram of an electronic device **300** according to an embodiment of the disclosure.

[0061] Referring to FIG. **3**, the electronic device **300** may be a device that tracks a subject based on a view area range designated by a user within a maximum view area obtainable through a camera, and provides an image, and may include a camera module **310**, a display **320**, at least one processor **330**, or memory **340**. The electronic device **300** in FIG. **3** may correspond to the electronic device **101** illustrated in FIG. **1**.

[0062] In an embodiment, the camera module **310** (for example, the camera module **180** of FIG. **1**) may shoot a subject according to an operation of a user. For example, the camera module **310** may acquire an image that includes the subject within a maximum view area obtainable by the camera module **310**. In various embodiments of the disclosure, the view area may be an area that is seen by the camera module **310**, and may correspond to a field of view. According to various embodiments, the electronic device **300** may include one or more cameras having different fields of view.

[0063] In an embodiment, the display **320** (for example, the display module **160** of FIG. **1**) may display an image which is acquired by the camera module **310**. For example, the display **320** may display an image corresponding to a view area that is set for the camera module **310**.

[0064] In an embodiment, the display **320** may be configured by one or more of a liquid crystal display (LCD), a thin film transistor LCD (TFT-LCD), organic light emitting diodes (OLED), a light emitting diode (LED), an active matrix organic LED (AMOLED), a flexible display and a 3 dimensional display. In addition, some of these displays may be configured in a transparent type or a light transmission type to allow a user to see the outside therethrough. This may be configured in a transparent display type including a transparent OLED (TOLED).

[0065] In an embodiment, the memory **340** (for example, the memory **130** of FIG. **1**) may store instructions that, when executed, control the at least one processor **330** (for example, the processor **120** of FIG. **1**) to perform various operations. For example, the at least one processor **330** may

perform operations of acquiring an image of a composition that is determined based on a view area range and an object area which are designated by the user when the auto-framing function of the camera is used.

[0066] In an embodiment, the at least one processor **330** may identify that the auto-framing function is activated while a camera application installed in the electronic device **300** is being executed. The at least one processor **330** may identify whether the auto-framing function is activated, based on configuration information on the camera module **310**. According to various embodiments, the at least one processor **330** may display a preview image on the display **320**, based on an initial zoom magnification of the camera module **310** when the camera application is executed. The initial zoom magnification of the camera module **310** may be a default zoom magnification (for example, 1.0 magnification) set by the camera application or a magnification predesignated by the user. The at least one processor **330** may maintain the default zoom magnification until a setting value related to a photographing composition is separately inputted by the user.

[0067] In an embodiment, when the at least one processor **330** identifies that the auto-framing function is activated, the at least one processor **330** may identify a maximum view area provided by the camera module **310**. The maximum view area may correspond to a maximum field of view that may be provided through the camera module **310**.

[0068] In an embodiment, the at least one processor **330** may acquire a first user input for designating a movable view area range within the maximum view area. The view area range may be understood as a range in which an object is detected within the maximum view area while photographing is being performed by using the camera module **310**. The at least one processor **330** may detect and continuously track an object while moving within the view area range designated based on the first user input. According to various embodiments, the at least one processor **330** may recommend a view area range that may be designated within the maximum view area to the user. The at least one processor **330** may detect one or more objects from a plurality of continuous frames which are acquired for a designated time by using the camera module **310**, and may determine a candidate view area range based on analysis on the one or more objects. For example, the at least one processor **330** may determine the candidate view area range based on at least one of the number, positions, motions, or directions of the one or more objects, and may display and recommend the determined candidate view area range on the display **320**. In this case, the at least one processor **330** may determine the candidate view area range by applying a margin of a ratio that is designated with reference to an area and a direction in which the motion of the at least one object is detected. The margin may be applied differently according to characteristics of the at least one object detected within the maximum view area, and may be adjusted to maintain the photographing composition (for example, a photographing angle or a zoom magnification) set by the user within the maximum view area. According to various embodiments, when a plurality of objects are detected within the maximum view area, the at least one processor **330** may determine the candidate view area range with reference to a direction of an object that has active motion or high importance among the plurality of objects. In another example, the at least one processor **330** may determine the candidate view area range to include all of the plurality of objects when the identified degrees of motion or importance of the plurality of objects are within the same or similar range.

[0069] In an embodiment, the at least one processor **330** may induce the first user input to reflect a composition desired by the user when the auto-framing function is used. For example, the at least one processor **330** may provide a first user interface to designate at least part of the maximum view area as the view area range. The first user interface may have a window box shape to set (or change) a range in horizontal directions and/or vertical directions within the maximum view area. The at least one processor **330** may determine the view area range based on the first user input which is acquired through the first user interface. According to various embodiments, the at least

one processor **330** may provide the first user interface based on the candidate view area range that is determined based on the analysis of the frame image. For example, the at least one processor **330** may arrange the first user interface on a position corresponding to the candidate area range. According to various embodiments, when it is identified that the view area range is not designated within the maximum view area by the user, the at least one processor **330** may automatically designate the candidate view area range determined based on the analysis of the frame image as the view area range.

[0070] In an embodiment, the at least one processor **330** may acquire a second user input for designating an object area to track within the view area range. The object area may be an area that includes a target object to be continuously tracked within the view area range, and may be designated to include a peripheral part of the object in addition to the face or upper body of the object. According to various embodiments, the at least one processor **330** may recommend an object area that may be designated within the view area range to the user. The at least one processor **330** may determine at least one candidate object area, based on information on one or more objects detected within the view area range, or user preference information. For example, the at least one processor **330** may determine, as a main object, an object that is designated by the user or an object that has a high frequency of appearance in images pre-stored in the memory **340**, and may designate the at least one candidate object area with reference to the main object. In another example, the at least one processor **330** may determine the main object, based on a type or a position of each of the detected one or more objects. For example, the at least one processor **330** may determine a person between the person and the animal as the main object, and, when a plurality of persons are detected, may determine, as the main object, a person positioned at the front (that is, a person closer to the electronic device **300**), or a person whose motion is continuously detected. The at least one processor **330** may designate the at least one candidate object area to include the determined main object.

[0071] In an embodiment, the at least one processor **330** may induce the second user input to reflect a composition of a subject desired by the user when the auto-framing function is used. For example, the at least one processor **330** may provide a second user interface for setting a boundary of the object area within the view area range. The second user interface may have a window box shape to set (or change) a range in horizontal directions and/or vertical directions with reference to the object detected within the view area range. The at least one processor **330** may determine the object area based on the second user input which is acquired through the second user interface. According to various embodiments, the at least one processor **330** may provide the second user interface based on the at least one candidate object area which is determined based on information on one or more objects within the view area range or user preference information. For example, the at least one processor **330** may arrange and provide the second user interface on a position corresponding to the at least one candidate object area. According to various embodiments, when it is identified that the object area is not designated within the view area range by the user, the at least one processor **330** may automatically designate the at least one candidate object area as the object area.

[0072] In an embodiment, the at least one processor **330** may determine a first view area to display on the display, based on the view area range which is designated by the first user input and the object area which is designated by the second user input. The at least one processor **330** may determine the first view area to include the object area within the view area range. According to various embodiments, the at least one processor **330** may determine the first view area by considering a zoom magnification on the object area. For example, the at least one processor **330** may calculate a minimum zoom magnification based on the size of the maximum view area and the size of the view area range. In this case, the at least one processor **330** may calculate the minimum zoom magnification by calculating a great value among values obtained by dividing the horizontal, vertical lengths of the maximum view area by the horizontal, vertical lengths of the view area

range, and may determine the first view area to make the object area seen at a zoom magnification greater than or less than the calculated minimum zoom magnification. In another example, the at least one processor **330** may acquire a magnification setting input including at least one of a minimum zoom magnification or maximum zoom magnification from the user. In this case, the at least one processor **330** may determine the first view area to maintain the maximum zoom magnification or lower and/or the minimum zoom magnification or higher for the object area within the maximum view area based on the magnification setting input. According to various embodiments, when it is identified that a plurality of object areas are designated by the second user input, the at least one processor **330** may determine the first view area to include all of the plurality of object areas. In this case, the first view area may be determined to maintain the minimum zoom magnification for the plurality of object areas.

[0073] In an embodiment, the at least one processor **330** may display an image corresponding to the determined first view area on the display **320**. The at least one processor **330** may provide the image corresponding to the first view area as a preview image of the running camera application, or may provide as a moving image screen.

[0074] In an embodiment, when it is identified that the object area is not included in the first view area while the image corresponding to the first view area is being provided through the display **320**, the at least one processor **330** may detect the object area again within the view area range. The at least one processor **330** may change the first view area based on the detected object area, and may display an image corresponding to the changed first view area on the display **320**.

[0075] In an embodiment, the at least one processor **330** may detect a third user input for changing the object area while providing the image corresponding to the first view area. In this case, the at least one processor **330** may determine a second view area based on the object area changed by the third user input, and may replace the image corresponding to the first view area with an image corresponding to the second view area and may provide the image through the display **320**.

[0076] FIGS. **4A**, **4B**, **4C**, and **4D** are views to explain a method for determining and displaying a view area when an auto-focusing function is activated according to various embodiments of the disclosure.

[0077] According to various embodiments, the electronic device **300** may provide an image based on a view area range and an object area which are designated by the user while photographing by using a camera module (for example, the camera module **180** of FIG. **1** or the camera module **310** of FIG. **3**).

[0078] Referring to FIG. **4A**, when it is detected that the auto-framing function of the camera module **310** is activated, the electronic device **300** may identify a maximum view area **400** which may be provided through the camera module **310**. The maximum view area **400** may be determined in response to a maximum field of view of the camera module **310**.

[0079] Referring to FIG. **4B**, the electronic device **300** may designate a movable view area range **410** to detect and track at least one object within the maximum view area **400**. The electronic device **300** may provide a user interface of a window box shape to designate the view area range **410**. For example, the electronic device **300** may designate the view area range **410** with reference four corner points **411**, **412**, **413**, **414** of the user interface which is set or moved by a user's touch input. Referring to FIG. **4B**, an area within a boundary that is formed with reference to the upper left corner point **411**, the lower left corner point **412**, the upper right corner point **413**, or the lower right corner point **414** of the user interface may be determined as the view area range **410**.

[0080] According to various embodiments, the electronic device **300** may calculate the view area range **410** that may be designated within the maximum view area **400**, and may recommend the same to the user. The electronic device **300** may analyze continuous frames which are obtained for a designated time, and may sort a background with a fixed position without movement, and objects with movement within the maximum view area **400**, and may determine a candidate view area range by considering the number, positions, motions or directions of the objects. For example,

when a plurality of objects are detected within the maximum view area **400**, the electronic device **300** may determine the candidate view area range to include all of the plurality of objects. The electronic device **300** may recommend the determined candidate view area range as the view area range **410**. In an embodiment, the electronic device **300** may determine the candidate view area range by applying a margin of a ratio that is designated with reference to an area and a direction in which motion of the object detected within the maximum view area **400** is detected. In this case, the electronic device **300** may apply a margin of a different ratio according to characteristics of the detected object. For example, when the detected object is a whole body of a specific person, the electronic device **300** may determine the candidate view area range to include a margin of 30% of the height of the whole body of the object in the upward direction, a margin of 10% of the height of the whole body in the downward direction, and a margin of 50% of the width of the whole body in the leftward and rightward directions, respectively. In another example, when a plurality of persons are detected as the object, the electronic device **300** may set different margins in the leftward/rightward/upward/downward directions according to motion areas and directions of the persons, and may determine the candidate view area range. According to various embodiments, in FIG. **4B**, the electronic device **300** may provide a user interface for designating the view area range **410** based on the determined candidate view area range. In this case, the electronic device **300** may set a position corresponding to the candidate view area range to an initial position of the user interface for designating the view area range **410**, and may display the user interface.

[0081] Referring to FIG. **4C**, the electronic device **300** may designate an object area **420** to track within the view area range. The electronic device **300** may provide a user interface of a window box shape to designate the object area **420**. The user interface for designating the object area **420** may be provided in the same way as that the user interface for designating the view area range **410** is provided, and an area within a boundary in the leftward/rightward/upward/downward directions that is formed by a user's touch input may be determined as the object area **420**.

[0082] According to various embodiments, the electronic device **300** may recommend the object area **420** that may be designated within the view area range to the user. The electronic device **300** may determine at least one candidate object area, based on information on one or more objects detected within the view area range **410**, or user preference information. For example, when it is identified that there is an object that is designated by the user or there is an object that appears in images pre-stored in memory (for example, the memory **130** of FIG. **1** or the memory **340** of FIG. **3**) more than a designated number of times, the electronic device **300** may designate the at least one candidate object area with reference to the corresponding object. In another example, the electronic device **300** may designate the candidate object area by considering an object selection priority. The object selection priority may be determined according to a type or a position of each of the one or more objects detected within the view area range **410**. The electronic device **300** may determine persons between persons and animals to have a relatively high priority, and, when a plurality of persons are detected, the electronic device **300** may determine a person positioned at the front to have a relatively high priority. The electronic device **300** may recommend the determined candidate object area as the object area **420**. According to various embodiments, in FIG. **4C**, the electronic device **300** may provide a user interface for designating the object area **420** based on the determined candidate object area. In this case, the electronic device **300** may set a position corresponding to the candidate object area to an initial position of the user interface for designating the object area **420**, and may display the user interface. The object area **420** may be designated to include a peripheral part of the object in addition to the face or upper body of the object to track within the view area range.

[0083] Referring to FIG. **4C**, the electronic device **300** may determine a view area **430** that is seen through a display (for example, the display module **160** of FIG. **1** or the display **320** of FIG. **3**) based on the designated view area range **410** and the designated object area **420**. The electronic device **300** may determine the view area **430** to include the object area **420** within the view area

range **410**.

[0084] Referring to FIG. **4D**, the electronic device **300** may display an image corresponding to the determined view area **420** on the display **320**. The image corresponding to the view area **430** may be provided as a preview image or a moving image screen while the camera module **310** is being executed. The electronic device **300** may be set to maintain a maximum zoom magnification for the object area **420** while providing the image corresponding to the view area **430**. In an embodiment, the minimum zoom magnification may be calculated by using the size of the maximum view area **400** and the size of the view area range **410**. For example, the electronic device **300** may determine, as the minimum zoom magnification of the view area **430**, a great value among the value obtained by dividing the horizontal length or width of the maximum view area **400** by the horizontal length or width of the view area range **410**, and the value obtained by dividing the vertical length or height of the maximum view area **400** by the vertical length or height of the view area range **410**. In an embodiment, the electronic device **300** may set the zoom magnification of the view area **430** based on magnification configuration information inputted by the user. In this case, the electronic device **300** may identify a maximum zoom magnification and/or a minimum zoom magnification which is designated by the user based on the magnification configuration information, and may set the zoom magnification on the object area **420** within the view area **430** to be maintained less than or equal to the maximum zoom magnification and/or greater than or equal to the minimum zoom magnification.

[0085] According to various embodiments, the electronic device **300** may continuously track the object area **420** while providing the image corresponding to the view area **430**. When it is identified that the object area **420** deviates from the view area **430**, the electronic device **300** may detect the object area **420** within the view area range **410** again, and may change and display the view area **430** based on the detected object area **420**.

[0086] According to various embodiments, the electronic device **300** may detect that the object area **420** is changed by a user's touch input while providing the image corresponding to the view area **430**. The electronic device **300** may determine a new view area **430** based on the changed object area **420**, and may provide an image corresponding to the new view area **430** on the display **320**.

[0087] FIG. **5** is a flowchart illustrating an operation method of an electronic device according to an embodiment of the disclosure.

[0088] According to an embodiment, the electronic device **300** may be a device that tracks a subject based on a view area range which is designated by a user within a maximum view area which may be acquired through a camera, and provides an image, and may correspond to the electronic device **101** shown in FIG. **1**. Operations of FIG. **5** may be performed by at least one processor (for example, the processor **120** of FIG. **1** or the at least one processor **330** of FIG. **3**) included in the electronic device **300**.

[0089] Referring to FIG. **5**, in operation **510**, the electronic device **300** may identify that an auto-framing function is activated while a camera application is being executed. The electronic device **300** may identify whether the auto-framing function is activated, based on configuration information on a camera module (for example, the camera module **180** of FIG. **1** or the camera module **310** of FIG. **3**).

[0090] According to an embodiment, in operation **520**, in response to identifying that the auto-framing function is activated, the electronic device **300** may identify a maximum view area provided by the camera module **310**. The maximum view area may correspond to a maximum field of view that may be provided through the camera module **310**.

[0091] According to an embodiment, in operation **530**, the electronic device **300** may acquire a first user input for designating a movable view area range within the maximum view area. The view area range may be understood as a range in which an object is detected within the maximum view area while photographing is being performed by using the camera module **310**. The electronic device **300** may detect and continuously track an object while moving within the view area range

designated based on the first user input. According to various embodiments, in operation 530, the electronic device 300 may recommend a view area range that may be designated within the maximum view area to the user. The electronic device 300 may determine a candidate view area range based on a result of analyzing a plurality of continuous frames which are acquired for a designated time through the camera module 310. For example, the electronic device 300 may detect one or more objects from the continuous frames, may determine the candidate view area range based on at least one of the number, positions, motions, or directions of the one or more objects, and may display the determined candidate view area range on a display (for example, the display module 160 of FIG. 1 or the display module 320 of FIG. 3). In this case, the electronic device 300 may determine the candidate view area range by applying a margin of a ratio that is designated with reference to an area and a direction in which the motion of the at least one object is detected. The margin may be applied differently according to characteristics of the at least one object detected within the maximum view area, and may be adjusted to maintain a photographing composition (for example, a photographing angle or a zoom magnification) set by the user within the maximum view area. According to various embodiments, when a plurality of objects are detected within the maximum view area, the electronic device 300 may determine the candidate view area range with reference to a direction of an object that has active motion or high importance among the plurality of objects. In another example, the at least one processor 330 may determine the candidate view area range to include all of the plurality of objects when the identified degrees of motion or importance of the plurality of objects are within the same or similar range. According to various embodiments, the electronic device 300 may provide a first user interface for designating at least part of the maximum view area as the view area range in order to induce the first user input. The first user interface may have a window box shape to set (or change) a range in horizontal directions and/or vertical directions within the maximum view area. The electronic device 300 may determine the view area range based on the first user input which is acquired through the first user interface. In an embodiment, the electronic device 300 may arrange and provide the first user interface on a position corresponding to the candidate area range which is determined based on analysis of the frame image. According to various embodiments, in operation 530, when it is identified that the view area range is not designated within the maximum view area by the user, the electronic device 300 may automatically designate the candidate view area range determined based on the analysis of the frame image as the view area range.

[0092] According to an embodiment, in operation 540, the electronic device 300 may acquire a second user input for designating an object area to track within the view area range. The object area may be an area that includes a target object to be continuously tracked within the view area range, and may be designated to include a peripheral part of the object in addition to the face or upper body of the object. According to various embodiments, in operation 540, the electronic device 300 may recommend an object area that may be designated within the view area range to the user. The electronic device 300 may determine at least one candidate object area, based on information on one or more objects detected within the view area range, or user preference information. For example, the electronic device 300 may determine, as a main object, an object that is designated by the user or an object that has a high frequency of appearance in images pre-stored in the memory 340, and may designate the at least one candidate object area with reference to the main object. In another example, the electronic device 300 may determine the main object, based on a type or a position of each of the detected one or more objects. In this case, the electronic device 300 may determine a person between the person and the animal as the main object, and, when a plurality of persons are detected, may determine, as the main object, a person positioned at the front (that is, a person closer to the electronic device 300), or a person whose motion is continuously detected. The electronic device 300 may designate the at least one candidate object area to include the determined main object. According to various embodiments, the electronic device 300 may provide a second user interface for setting a boundary of the object area within the view area range in order to induce

the second user input. The second user interface may have a window box shape to set (or change) a range in horizontal directions and/or vertical directions within the view area range. The electronic device **300** may determine the object area based on the second user input which is acquired through the second user interface. In an embodiment, the electronic device **300** may arrange and provide the second user interface on a position corresponding to the candidate object area which is determined based on information on one or more objects within the view area range or user preference information. According to various embodiments, in operation **540**, when it is identified that the object area is not designated within the view area range by the user, the electronic device **300** may automatically designate the candidate object area as the object area.

[0093] According to an embodiment, in operation **550**, the electronic device **300** may determine a first view area to display on the display, based on the view area range which is designated by the first user input and the object area which is designated by the second user input. The electronic device **300** may determine the first view area to include the object area within the view area range. According to various embodiments, in operation **550**, the electronic device **300** may determine the first view area by considering a zoom magnification on the object area. For example, the electronic device **300** may calculate a minimum zoom magnification based on the size of the maximum view area and the size of the view area range. In this case, the electronic device **300** may calculate the minimum zoom magnification by calculating a great value among values obtained by dividing the horizontal, vertical lengths of the maximum view area by the horizontal, vertical lengths of the view area range, and may determine the first view area to make the object area seen at a zoom magnification greater than or less than the calculated minimum zoom magnification. In another example, the electronic device **300** may acquire a magnification setting input including at least one of a minimum zoom magnification or maximum zoom magnification from the user. In this case, the electronic device **300** may determine the first view area to maintain the maximum zoom magnification or lower and/or the minimum zoom magnification or higher for the object area within the maximum view area based on the magnification setting input. According to various embodiments, when it is identified that a plurality of object areas are designated by the second user input, the electronic device **300** may determine the first view area to include all of the plurality of object areas. In this case, the first view area may be determined to maintain the minimum zoom magnification for the plurality of object areas.

[0094] According to an embodiment, in operation **560**, the electronic device **300** may display an image corresponding to the first view area on the display **320**. The electronic device **300** may provide the image corresponding to the first view area as a preview image of the running camera application, or may provide as a moving image screen. According to various embodiments, when it is identified that the object area is not included in the first view area while the image corresponding to the first view area is being provided, the electronic device **300** may detect the object area again within the view area range. The electronic device **300** may change the first view area based on the detected object area, and may display an image corresponding to the changed first view area on the display **320**.

[0095] According to an embodiment, in operation **560**, the electronic device **300** may detect a third user input for changing the object area while providing the image corresponding to the first view area. The electronic device **300** may determine a second view area based on the object area changed by the third user input, and may replace the image corresponding to the first view area with an image corresponding to the second view area and may provide the image through the display **320**.

[0096] FIG. **6** is a flowchart to explain detailed operations of determining a view area range by an electronic device according to an embodiment of the disclosure.

[0097] According to an embodiment, the electronic device **300** may be a device that determines a view area range which is movable for detecting and tracking an object within a maximum view area provided by using a camera, and may correspond to the electronic device **101** shown in FIG. **1**.

Operations of FIG. 6 may be performed by at least one processor (for example, the processor 120 of FIG. 1 or the at least one processor 330 of FIG. 3) included in the electronic device 300.

[0098] Referring to FIG. 6, in operation 610, the electronic device 300 may receive an input of a camera image. The electronic device 300 may display an image that is inputted through a camera module (for example, the camera module 180 of FIG. 1 or the camera module 310 of FIG. 3) on a display (for example, the display module 160 of FIG. 1 or the display 320 of FIG. 3) as a preview image.

[0099] According to an embodiment, in operation 620, the electronic device 300 may recommend a view area range that may be designated within a maximum view area, and a target object to be tracked. The electronic device 300 may determine a candidate view area range and a candidate object to recommend to the user by analyzing a plurality of continuous frames acquired for a designated time. According to various embodiments, in operation 620, the electronic device 300 may detect one or more objects whose motion is detected as a result of analyzing the plurality of continuous frames, and may determine the candidate object based on information on the one or more objects or user preference information. For example, when there is an object that is designated by the user or there is an object that appears in images pre-stored in memory (for example, the memory 130 of FIG. 1 or the memory 340 of FIG. 3) more than a designated number of times, the electronic device 300 may determine the corresponding object as the candidate object. In another example, when it is identified that there is no object that is designated by the user or there is no object that appears more than the designated number of times, the electronic device 300 may determine the candidate object by considering a type or a position of each of the one or more objects. The electronic device 300 may determine a person between the person and the animal as the candidate object, and, when a plurality of persons are detected, the electronic device 300 may determine a person positioned at the front (that is, a person closer to the electronic device 300) as the candidate object. According to various embodiments, in operation 620, the electronic device 300 may determine the candidate view area range based on at least one of the number, positions, motions or directions of the one or more detected objects. The electronic device 300 may determine the candidate view area range by applying a margin of a ratio which is designated with reference to an area and a direction in which the motion of the at least one object is detected. The margin may be applied differently according to characteristics of the at least one object detected within the maximum view area. The electronic device 300 may display the determined candidate object and the candidate view area range on the display 320, and may recommend as the object to track and the view area range.

[0100] According to an embodiment, in operation 630, the electronic device 300 may determine whether a view area range is set within the maximum view area by the user. The electronic device 300 may identify whether the view area range is set according to whether a touch input of the user for designating the view area range within the maximum view area is detected.

[0101] When the view area range is set by the user as a result of identifying in operation 630 (operation 630-Yes), the electronic device 300 may calculate a minimum zoom magnification in operation 640. For example, the electronic device 300 may calculate a great value among values that are obtained by dividing horizontal, vertical lengths of the maximum view area by horizontal, vertical lengths of the view area range, as the minimum zoom magnification. In another example, the electronic device 300 may identify at least one of a minimum zoom magnification or a maximum zoom magnification based on magnification configuration information inputted by the user. In operation 645, the electronic device 300 may determine a first view area including the object to track within the set view area range. The electronic device 300 may determine the first view area to make the object seen at a zoom magnification greater than or equal to the calculated minimum magnification, or may determine the first view area to maintain a zoom magnification between the identified minimum zoom magnification and maximum zoom magnification.

[0102] When the view area range is not set by the user as a result of identifying in operation 630

(operation **630-No**), the electronic device **300** may determine a certain area that includes the object to track within the maximum view area as the first view area in operation **650**. According to various embodiments, in operation **650**, the electronic device **300** may determine the first view area to include the object based on the candidate view area range.

[0103] According to an embodiment, in operation **660**, the electronic device **300** may move a focus and/or an angle of the camera module **310** to the determined first view area, and may provide an image corresponding to the first view area via the display **320**.

[0104] According to an embodiment, in operation **670**, the electronic device **300** may identify whether the object is continuously detected within the view area range while providing the image corresponding to the first view area. For example, when it is detected that the object deviates from the view area range, the electronic device **300** may determine that the electronic device **300** fails to detect (or track) the object.

[0105] When it is identified that the electronic device fails to detect the object within the view area range as a result of identifying in operation **670** (operation **670-Yes**), the electronic device **300** may return to operation **645**.

[0106] When it is identified that the object is continuously detected within the view area range as a result of identifying in operation **670** (operation **670-No**), the electronic device **300** may display a preview image based on the first view area including the object in operation **680**. According to various embodiments, when it is detected that a shape or a position is changed due to motion of the object while the preview image is being displayed, the electronic device **300** may change the first view area based on the shape or position changed for the object, and may provide an image corresponding to the changed first view as the preview image.

[0107] According to an embodiment, in operation **690**, the electronic device **300** may identify whether the view area range is changed while providing the preview image corresponding to the first view area. When a user's touch input to change the view area range occurs or the target object to be tracked is changed, the electronic device **300** may determine that the view area range is changed. When the view area range is changed as a result of identifying in operation **690** (operation **690-Yes**), the electronic device may return to operation **630**. When the view area range is not changed as a result of identifying in operation **690** (operation **690-No**), the electronic device may return to operation **670**.

[0108] FIG. 7 is a flowchart illustrating detailed operations of determining an object area by an electronic device according to an embodiment of the disclosure.

[0109] According to an embodiment, the electronic device **300** may be a device that continuously determines an area of an object to track while acquiring an image by using a camera, and may correspond to the electronic device **101** shown in FIG. 1. Operations of FIG. 7 may be performed by at least one processor (for example, the processor **120** of FIG. 1 or the at least one processor **330** of FIG. 3) included in the electronic device **300**.

[0110] Referring to FIG. 7, in operation **710**, the electronic device **300** may receive an input of a camera image. The electronic device **300** may display an image inputted through a camera module (for example, the camera module **180** of FIG. 1 or the camera module **310** of FIG. 3) on a display (for example, the display module **160** of FIG. 1 or the display **320** of FIG. 3) as a preview image.

[0111] According to an embodiment, in operation **720**, the electronic device **300** may recommend a target object to track. The electronic device **300** may determine a candidate object to recommend to the user, based on a result of analyzing a plurality of continuous frames which are acquired for a designated time. According to various embodiments, in operation **720**, the electronic device **300** may sort a background with a fixed position without movement and one or more objects with movement from the plurality of continuous frames, and may determine the candidate object based on information on the sorted one or more objects or user preference information. For example, when there is an object that is designated by the user or there is an object that appears in images pre-stored in memory (for example, the memory **130** of FIG. 1 or the memory **340** of FIG. 3) more

than a designated number of times, the electronic device **300** may determine the corresponding object as the candidate object. In another example, when it is identified that there is no object that is designated by the user or there is no object that appears more than the designated number of times, the electronic device **300** may determine the candidate object by considering a type or a position of each of the one or more objects. The electronic device **300** may determine a person between the person and the animal as the candidate object, and, when a plurality of persons are detected, may determine a person positioned at the front (that is, a person closer to the electronic device **300**) as the candidate object.

[0112] According to an embodiment, in operation **730**, the electronic device **300** may determine whether an object area that is a target to be tracked is set by the user. The electronic device **300** may determine whether the object area is set according to whether a touch input of the user for designating the object area is detected within the maximum view area.

[0113] When the object area is not set by the user as a result of identifying in operation **730** (operation **730**-No), the electronic device **300** may automatically set a certain area including the object to track within the maximum view area as the object area in operation **735**. According to various embodiments, in operation **735**, the electronic device **300** may determine the object area with reference to the candidate object. The electronic device **300** may designate the object area by including a peripheral part of the object in addition to the face or upper body, the whole body of the candidate object. When the object area is set, the electronic device **300** may determine a first view area that is seen by the camera module **310** in operation **740**. The electronic device **300** may determine the first view area to include the object area, and may move the focus and/or angle of the camera module **310** to the determined first view area and may provide an image corresponding to the first view area through the display **320**.

[0114] When the object area is set by the user as a result of identifying in operation **730** (operation **730**-Yes), the electronic device **300** may determine the first view area based on the object area which is set by the user in operation **740**. The electronic device **300** may determine the first view area to include the object area set by the user, and may move the focus and/or angle of the camera module **310** to the determined first view area and may provide the image corresponding to the first view area through the display **320**.

[0115] According to an embodiment, in operation **750**, the electronic device **300** may determine whether the object area is continuously detected within the view area range while providing the image corresponding to the first view area. For example, when it is detected that the object area deviates from the view area range, the electronic device **300** may determine that the object area is not detected within the view area range.

[0116] When it is identified that the object area is not detected within the view area range as a result of identifying in operation **750** (operation **750**-No), the electronic device **300** may return to operation **740** to determine the first view area again.

[0117] When it is identified that the object area is detected within the view area range as a result of identifying in operation **750** (operation **750**-Yes), the electronic device **300** may provide the image corresponding to the first view area including the object area as a preview image in operation **760**.

[0118] FIGS. **8**, **9**, **10** and **11** are views to explain a method for providing an image corresponding to a view area determined based on an object area according to various embodiments of the disclosure.

[0119] According to various embodiments, the electronic device **300** may provide an image based on an object area designated by a user while photographing using a camera module (for example, the camera module **180** of FIG. **1** or the camera module **310** of FIG. **3**).

[0120] Referring to FIG. **8**, the electronic device **300** may determine an object area **810** to track in a maximum view area **800** which may be provided through the camera module. For example, the electronic device **300** may identify that the head and the upper body of a first object among a plurality of objects detected in the maximum view area are designated as the object area **810**. The

electronic device **300** may determine a view area **820** including the object area **810** by considering a size and/or a ratio of the object area **810** in order to provide an image of a composition giving stability. In FIG. **8**, the electronic device **300** may identify that the vertical length (or height) of the designated object area **810** has a long ratio compared to the horizontal length (or width), and may determine the view area **820** to apply a margin of a greater ratio to the horizontal direction of the object area **810** than to the vertical direction.

[0121] Referring to FIG. **9**, the electronic device **300** may determine an object area **910** to track in a maximum view area **900** which may be provided through the camera module. For example, the electronic device **300** may identify that the head of the first object among the plurality of objects detected in the maximum view area is designated as the object area **910**. In this case, the electronic device **300** may determine the view area **920** by applying a margin of a ratio which is set differently to the leftward/rightward/upward/downward directions of the head, respectively. The margin ratio may be pre-designated according to a type or a part of the object included in the object area **910**.

[0122] Referring to FIG. **10**, the electronic device **300** may determine an object area **1010** to track in a maximum view area **1000** which may be provided through the camera module. For example, the electronic device **300** may identify that the head of the first object among the plurality of objects detected in the maximum view area, and an upper margin of the head are designated as the object area **1010**. In this case, the electronic device **300** may determine the view area **1020** by applying a margin of a greater ratio to the downward direction of the object area **1010** than to the upward direction by considering the upper margin included in the object area **1010**.

[0123] Referring to FIG. **11**, the electronic device **300** may determine a plurality of object areas **1110**, **1115** to track in a maximum view area **1100** which may be provided through the camera module. For example, the electronic device **300** may identify that the head of the first object among the plurality of objects detected in the maximum view area is designated as a first object area **1110**, and a head of a second object is designated as a second object area **1115**. In this case, the electronic device **300** may determine a view area **1120** to include both the first object area **1110** and the second object area **1115**. According to various embodiments, the electronic device **300** may determine the view area **1120** by applying the margin of the same ratio to the rightward direction of the first object area **1110** and the leftward direction of the second object area **1115** in order to provide an image of a composition giving stability.

[0124] According to various embodiments, the electronic device **300** may determine the view area **820**, **920**, **1020**, **1120** to maintain a zoom magnification setting which is designated by the user or a minimum zoom magnification which is calculated based on the sizes of the maximum view area and the view area range while providing an image corresponding to the view area **820**, **920**, **1020**, **1120**.

[0125] Referring to FIGS. **8** to **11**, the electronic device **300** may determine the view area **820**, **920**, **1020**, **1120** that is shown to the user differently according to the sizes, ratios, number of the object areas **810**, **910**, **1010**, **1110**, **1115** designated by the user while photographing by using the camera module.

[0126] According to an embodiment, an electronic device (for example, the electronic device **300**) may include a camera module (for example, the camera module **310**), a display (for example, the display **320**), at least one processor (for example, the processor **330**) operatively connected with the camera module and the display, and memory (for example, the memory **340**) operatively connected with the at least one processor, and the memory may store instructions that, when executed, cause the at least one processor to: identify a maximum view area that is provided by the camera module in response to identifying that an auto-framing function is activated while a camera application is being executed; acquire a first user input for designating a view area range that is movable within the maximum view area; acquire a second user input for designating an object area to track within the view area range; determine a first view area comprising the object area based on the first user

input and the second user input; and display an image corresponding to the determined first view area on the display.

[0127] In an embodiment, the instructions may cause the at least one processor to: calculate a minimum zoom magnification on the first view area based on a size of the maximum view area and a size of the view area range; and determine the first view area based on the calculated minimum zoom magnification.

[0128] In an embodiment, the instructions may cause the at least one processor to: detect one or more objects from a plurality of continuous frames which are acquired for a designated time by using the camera module; determine a candidate view area range based on at least one of number, positions, motions or directions of the one or more objects; and display the determined candidate view area range on the display.

[0129] In an embodiment, the instructions may cause the at least one processor to: provide a first user interface to designate at least part of the maximum view area as the view area range, based on the candidate view area range; and determine the view area range based on the first user input which is acquired through the first user interface.

[0130] In an embodiment, the instructions may cause the at least one processor to: determine at least one candidate object area based on information on the one or more objects or user preference information; and to display the determined at least one candidate object area on the display.

[0131] In an embodiment, the instructions may cause the at least one processor to: provide a second user interface to set a boundary of the object area within the view area range, based on the determined at least one candidate object area; and determine the first view area based on the second user input which is acquired through the second user interface.

[0132] In an embodiment, the instructions may cause the at least one processor to: when it is identified that the object area is not included while displaying the image corresponding to the first view area on the display, detect the object area within the view area range; and change the first view area based on the detected object area, and display the changed first view area on the display.

[0133] In an embodiment, the instructions may cause the at least one processor to: acquire a third user input to change the object area within the view area range; determine a second view area based on the object area changed by the third user input; and display an image corresponding to the second view area on the display.

[0134] In an embodiment, the instructions may cause the at least one processor to, when it is identified that a plurality of object areas are designated by the second user input, determine the first view area to include all of the plurality of object areas.

[0135] In an embodiment, the instructions may cause the at least one processor to: acquire a magnification setting input including at least one of a minimum zoom magnification or a maximum zoom magnification from the user; and determine the first view area within the maximum view area based on the magnification setting input.

[0136] According to an embodiment, an operation method of an electronic device (for example, the electronic device **300**) may include: identifying a maximum view area that is provided by a camera module of the electronic device in response to identifying that an auto-framing function is activated while a camera application is being executed; acquiring a first user input for designating a view area range that is movable within the maximum view area; acquiring a second user input for designating an object area to track within the view area range; determining a first view area including the object area based on the first user input and the second user input; and displaying an image corresponding to the determined first view area.

[0137] In an embodiment, determining the first view area may include: calculating a minimum zoom magnification on the first view area based on a size of the maximum view area and a size of the view area range; and determining the first view area based on the calculated minimum zoom magnification.

[0138] In an embodiment, acquiring the first user input may include: detecting one or more objects

from a plurality of continuous frames which are acquired for a designated time by using the camera module; determining a candidate view area range based on at least one of number, positions, motions or directions of the one or more objects; and displaying the determined candidate view area range on a display of the electronic device.

[0139] In an embodiment, the operation method may further include: providing a first user interface to designate at least part of the maximum view area as the view area range, based on the candidate view area range; and determining the view area range based on the first user input which is acquired through the first user interface.

[0140] In an embodiment, acquiring the second user input may include: determining at least one candidate object area based on information on the one or more objects or user preference information, and displaying the determined at least one candidate object area on the display of the electronic device.

[0141] In an embodiment, the operation method may further include: providing a second user interface to set a boundary of the object area within the view area range, based on the at least one candidate object area; and determining the first view area based on the second user input which is acquired through the second user interface.

[0142] In an embodiment, determining the first view area may include: when it is identified that the object area is not included in the image while displaying the image corresponding to the first view area, detecting the object area within the view area range; and changing the first view area based on the detected object area, and displaying the changed first view area.

[0143] In an embodiment, the operation method may further include: acquiring a third user input to change the object area within the view area range; determining a second view area based on the object area changed by the third user input; and displaying an image corresponding to the second view area.

[0144] In an embodiment, determining the first view area may include, when it is identified that a plurality of object areas are designated by the second user input, determining the first view area to include all of the plurality of object areas.

[0145] In an embodiment, determining the first view area may include: acquiring a magnification setting input including at least one of a minimum zoom magnification or a maximum zoom magnification from the user; and determining the first view area within the maximum view area based on the magnification setting input.

[0146] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0147] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another

element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0148] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0149] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0150] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0151] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0152] It will be appreciated that various embodiments of the disclosure according to the claims and description in the specification can be realized in the form of hardware, software or a combination of hardware and software.

[0153] Any such software may be stored in non-transitory computer readable storage media. The non-transitory computer readable storage media store one or more computer programs (software modules), the one or more computer programs include computer-executable instructions that, when executed by one or more processors of an electronic device individually or collectively, cause the electronic device to perform a method of the disclosure.

[0154] Any such software may be stored in the form of volatile or non-volatile storage such as, for example, a storage device like read only memory (ROM), whether erasable or rewritable or not, or in the form of memory such as, for example, random access memory (RAM), memory chips, device or integrated circuits or on an optically or magnetically readable medium such as, for example, a compact disk (CD), digital versatile disc (DVD), magnetic disk or magnetic tape or the like. It will be appreciated that the storage devices and storage media are various embodiments of non-transitory machine-readable storage that are suitable for storing a computer program or computer programs comprising instructions that, when executed, implement various embodiments of the disclosure. Accordingly, various embodiments provide a program comprising code for implementing apparatus or a method as claimed in any one of the claims of this specification and a non-transitory machine-readable storage storing such a program.

[0155] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: a camera module; a display; memory, comprising one or more storage media, storing instructions; and at least one processor operatively connected with the camera module, the display, and the memory, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: identify a maximum view area that is provided by the camera module in response to identifying that an auto-framing function is activated while a camera application is being executed, acquire a first user input for designating a view area range that is movable within the maximum view area, acquire a second user input for designating an object area to track within the view area range, determine a first view area comprising the object area based on the first user input and the second user input, and display an image corresponding to the determined first view area on the display.
2. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors individually or collectively, further cause the electronic device to: calculate a minimum zoom magnification on the first view area based on a size of the maximum view area and a size of the view area range; and determine the first view area based on the calculated minimum zoom magnification.
3. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors individually or collectively, further cause the electronic device to: detect one or more objects from a plurality of continuous frames which are acquired for a designated time by using the camera module; determine a candidate view area range based on at least one of number, positions, motions or directions of the one or more objects; and display the determined candidate view area range on the display.
4. The electronic device of claim 3, wherein the instructions, when executed by the one or more processors individually or collectively, further cause the electronic device to: provide a first user interface to designate at least part of the maximum view area as the view area range, based on the candidate view area range; and determine the view area range based on the first user input which is acquired through the first user interface.
5. The electronic device of claim 3, wherein the instructions, when executed by the one or more processors individually or collectively, further cause the electronic device to: determine at least one candidate object area based on information on the one or more objects or user preference information; provide a second user interface to set a boundary of the object area within the view area range, based on the determined at least one candidate object area; and determine the first view area based on the second user input which is acquired through the second user interface.

- 6.** The electronic device of claim 1, wherein the instructions, when executed by the one or more processors individually or collectively, further cause the electronic device to: when it is identified that the object area is not included while displaying the image corresponding to the first view area on the display, detect the object area within the view area range; and change the first view area based on the detected object area, and display the changed first view area on the display.
- 7.** The electronic device of claim 1, wherein the instructions, when executed by the one or more processors individually or collectively, further cause the electronic device to: acquire a third user input to change the object area within the view area range; determine a second view area based on the object area changed by the third user input; and display an image corresponding to the second view area on the display.
- 8.** The electronic device of claim 1, wherein the instructions, when executed by the one or more processors individually or collectively, further cause the electronic device to, when it is identified that a plurality of object areas are designated by the second user input, determine the first view area to comprise all of the plurality of object areas.
- 9.** The electronic device of claim 1, wherein the instructions, when executed by the one or more processors individually or collectively, further cause the electronic device to: acquire a magnification setting input comprising at least one of a minimum zoom magnification or a maximum zoom magnification from the user; and determine the first view area within the maximum view area based on the magnification setting input.
- 10.** An operation method of an electronic device, the operation method comprising: identifying a maximum view area that is provided by a camera module of the electronic device in response to identifying that an auto-framing function is activated while a camera application is being executed; acquiring a first user input for designating a view area range that is movable within the maximum view area; acquiring a second user input for designating an object area to track within the view area range; determining a first view area comprising the object area based on the first user input and the second user input; and displaying an image corresponding to the determined first view area.
- 11.** The operation method of claim 10, wherein determining the first view area comprises: calculating a minimum zoom magnification on the first view area based on a size of the maximum view area and a size of the view area range; and determining the first view area based on the calculated minimum zoom magnification.
- 12.** The operation method of claim 10, wherein acquiring the first user input comprises: detecting one or more objects from a plurality of continuous frames which are acquired for a designated time by using the camera module; determining a candidate view area range based on at least one of number, positions, motions or directions of the one or more objects; and displaying the determined candidate view area range on a display of the electronic device.
- 13.** The operation method of claim 10, wherein determining the first view area comprises: when it is identified that the object area is not included in the image while displaying the image corresponding to the first view area, detecting the object area within the view area range; and changing the first view area based on the detected object area, and displaying the changed first view area.
- 14.** The operation method of claim 10, further comprising: acquiring a third user input to change the object area within the view area range; determining a second view area based on the object area changed by the third user input; and displaying an image corresponding to the second view area.
- 15.** The operation method of claim 10, wherein determining the first view area comprises, when it is identified that a plurality of object areas are designated by the second user input, determining the first view area to comprise all of the plurality of object areas.
- 16.** The method of claim 12, further comprising: providing a first user interface to designate at least part of the maximum view area as the view area range, based on the candidate view area range; and determining the view area range based on the first user input which is acquired through the first user interface.

17. The method of claim 12, further comprising: determining at least one candidate object area based on information on the one or more objects or user preference information; providing a second user interface to set a boundary of the object area within the view area range, based on the determined at least one candidate object area; and determining the first view area based on the second user input which is acquired through the second user interface.

18. The method of claim 10, further comprising: acquiring a magnification setting input comprising at least one of a minimum zoom magnification or a maximum zoom magnification from the user; and determining the first view area within the maximum view area based on the magnification setting input.

19. One or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by one or more processors of an electronic device individually or collectively, cause the electronic device to perform operations, the operations comprising: identifying a maximum view area that is provided by a camera module of the electronic device in response to identifying that an auto-framing function is activated while a camera application is being executed; acquiring a first user input for designating a view area range that is movable within the maximum view area; acquiring a second user input for designating an object area to track within the view area range; determining a first view area comprising the object area based on the first user input and the second user input; and displaying an image corresponding to the determined first view area.

20. The one or more non-transitory computer-readable storage media of claim 19, the operations further comprising: acquiring a third user input to change the object area within the view area range; determining a second view area based on the object area changed by the third user input; and displaying an image corresponding to the second view area.
